

markdown2html5-base

Full Markdown Functionality Reference

This reference covers all `markdown2html5-base` features.

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Programmatic Usage

```
from markdown2html5_base import MarkdownToHTML
converter = MarkdownToHTML()
html = converter.convert("# Hello")
print(html)
```

1. Headings (H1-H6)

Markdown	Output HTML
# Heading 1	Heading 1
## Heading 2	Heading 2
### Heading 3	Heading 3
#### Heading 4	Heading 4
##### Heading 5	Heading 5
##### Heading 6	Heading 6

Custom ID: ## Section {#sec1} => <h2 id="sec1">Section</h2>

2. Inline Text Styling

Markdown	Output HTML
<code>***bold italic***</code>	<code>bold italic</code>
<code>___bold italic___</code>	<code>bold italic</code>
<code>**bold**</code>	<code>bold</code>
<code>__bold__</code>	<code>bold</code>
<code>*italic*</code>	<code>italic</code>
<code>_italic_</code>	<code>italic</code>
<code>~~strikethrough~~</code>	<code><s>strikethrough</s></code>
<code>==highlight==</code>	<code><mark>highlight</mark></code>
<code>~subscript~</code>	<code><sub>subscript</sub></code>
<code>^^underline^^</code>	<code><u>underline</u></code>
<code>^superscript^</code>	<code><sup>superscript</sup></code>
<code>`code`</code>	<code><code>code</code></code>

3. Links and Images

Markdown	Output HTML
<code>[text](url)</code>	<code>text</code>
<code>![alt](src)</code>	<code><figure></figure></code>
<code>![alt](src "Title")</code>	<code><figure> <figcaption>Title</figcaption></figure></code>

4. Fenced Code Blocks

A language tag after the opening fence (three backticks before and after the code) is rendered as a `<div class="code-lang">/python/</div>` label above the `<code>` element:

```
<div class="code-lang">&sol;python&sol;</div><pre><code>def hello():  
    print("Hi")</code></pre>
```

Without a language tag, the code renders plainly as `<pre><code>` .

5. Horizontal Rules

--- / *** / ___ (3+ chars) => `<hr>`

6. Lists

1. **Unordered:** * Item or - Item => `<ul><li>Item</li></ul>`

2. **Ordered:** 1. Item => `<ol><li>Item</li></ol>`

3. **Task:**

- [] todo => `<ul><li><input type="checkbox" disabled> todo</li></ul>`

1. [x] done => `<ol><li><input type="checkbox" checked disabled> done</li></ol>`

4. **Nested:** indent an item with 2 (or 4) spaces or a tab to nest a list one level deeper. Nesting works to any depth, and each level may switch freely between bullets and numbers:

```
- Parent
  - Child
    1. Grandchild
```

=>

```
<ul>
  <li>Parent
    <ul>
      <li>Child
        <ol>
          <li>Grandchild</li>
        </ol>
      </li>
    </ul>
  </li>
</ul>
```

7. TOC (Table of Contents)

The Table of Contents (TOC) combines the styles of Heading 2, unordered lists, and links.

```
## Table of Contents {#toc}
```

- [1. Headings (H1-H6)](#1-headings-h1-h6)
- [2. Inline Text Styling](#2-inline-text-styling)
- [3. Links and Images](#3-links-and-images)

=>

```
<h2 id="toc">Table of Contents</h2>
<ul>
  <li><a href="#1-headings-h1-h6">1. Headings (H1-H6)</a></li>
  <li><a href="#2-inline-text-styling">2. Inline Text Styling</a></li>
  <li><a href="#3-links-and-images">3. Links and Images</a></li>
</ul>
```

8. Blockquotes

> text => <blockquote><p>text</p></blockquote> : blank lines within a blockquote split into separate <p> tags.

9. Tables

Supports <thead> , <tbody> , <tfoot> , and alignment:

Separator	Alignment
:---	left
:---:	center
---:	right

Footer: a row of = signs below the separator columns, after the body rows, renders a <tfoot> .

```
| Product | Qty | Price |
| :----- | :-: | ----: |
| Apples  | 2   | $3.00 |
| Bananas | 3   | $1.50 |
| Cherries | 1   | $4.00 |
| =====|=====|=====|
| Total   | 6   | $8.50 |
```

=>

```

<table>
  <thead>
    <tr>
      <th style="text-align:left;">Product</th>
      <th style="text-align:center;">Qty</th>
      <th style="text-align:right;">Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td style="text-align:left;">Apples</td>
      <td style="text-align:center;">2</td>
      <td style="text-align:right;">$3.00</td>
    </tr>
    <tr>
      <td style="text-align:left;">Bananas</td>
      <td style="text-align:center;">3</td>
      <td style="text-align:right;">$1.50</td>
    </tr>
    <tr>
      <td style="text-align:left;">Cherries</td>
      <td style="text-align:center;">1</td>
      <td style="text-align:right;">$4.00</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td style="text-align:left;">Total</td>
      <td style="text-align:center;">6</td>
      <td style="text-align:right;">$8.50</td>
    </tr>
  </tfoot>
</table>

```

10. Definition Lists

Term

: Definition 1

: Definition 2

=>

```
<dl>
  <dt>Term</dt>
  <dd>Definition 1</dd>
  <dd>Definition 2</dd>
</dl>
```

11. Footnotes

Reference: `[^1]` => `<sup id="fnref:1"><a href="#fn:1" class="footnote-ref">1</a></sup>`

Definition: `[^1]: Text` at bottom => rendered in `<div class="footnotes"><ol>...</ol></div>`

12. Language Markers

Annotate blocks or inline text with a language using a valid BCP 47 tag (e.g. `de`, `fr`, `zh-Hans`).

- **Block level:** Place `{:lang}` (with a space after it) at the very start of a line, before the Markdown tag. It renders as the global `lang` attribute on the resulting element and needs no closing marker:

```
{:de} # Überschrift
{:fr} Un paragraphe français.
{:ru} - Пункт списка
{:de} > Ein Zitat
{:de} | Kopf | Kopf |
      | ---- | ---- |
      | Zelle | Zelle |
```

=>

```
<h1 lang="de">Überschrift</h1>
<p lang="fr">Un paragraphe français.</p>
<ul lang="ru">
  <li>Пункт списка</li>
</ul>
<blockquote lang="de">...</blockquote>
<table lang="de">...</table>
```

- **Inline level:** Wrap text with `{:lang}...{:}` to render a `<span lang="...">:`

A French phrase `{:fr}"L'État c'est moi"{:}` is traditionally attributed to King Louis XIV of France.

=>

`<p>A French phrase <span lang="fr">"L'État c'est moi"</span> is traditionally attributed to King Louis XIV of France</p>`

13. Ruby Annotations

`{日本語|にほんご}` => `<ruby>日本語<rp>(</rp><rt>にほんご</rt><rp>)</rp></ruby>`

14. Emoji Shortcodes

- `:joy:` 😂
- `:smile:` 😊
- `:heart:` ❤️
- `:thumbsup:` 👍
- `:thumbsdown:` 👎
- `:wink:` 😉
- `:tada:` 🎉
- `:rocket:` 🚀
- `:fire:` 🔥
- `:star:` ★
- `:cry:` 😭
- `:thinking:` 🤔
- `:100:` 100
- `:sparkles:` ✨
- `:eyes:` 👁️
- `:bulb:` 💡
- `:warning:` ⚠️
- `:ok:` 🏆
- `:check_mark:` ✓

15. Typography and Legal Marks

Name	Input	HTML Output
Copyright	(c)	©
Trademark	(tm)	™
Registered Trademark	(r)	®
Plus-Minus	+/-	±
Not Equal To	!=	≠
Logical Equivalence	<=>	⇔
Less-Than or Equal To	<=	≤
Greater-Than or Equal To	>=	≥
Right Arrow	->	→
Left Arrow	<-	←
Up Arrow	↑	↑
Down Arrow	↓	↓
Logical Implication	=>	⇒
Solidus (Slash)	/	/
Reverse Solidus (Backslash)	\	\
One-Half	1/2	½
One-Third	1/3	⅓
Two-Thirds	2/3	⅔
One-Quarter	1/4	¼
Three-Quarters	3/4	¾
Left Angle Quote	<<	«
Right Angle Quote	>>	»
Left Double Quote	“	“
Right Double Quote	”	”
Smart Double Quotes	"text"	“text”

Name	Input	HTML Output
Smart Single Quotes	'text'	‘text’
Straight Apostrophe	'	'
Em Dash	---	—
En Dash	--	–
Ellipsis	...	…
Non-Breaking Space	 	

16. Hard Line Breaks

End line with two spaces or backslash: `<br />`

17. HTML Comments

[comment]: # => `<!--comment-->`

18. YAML Front Matter

If the file begins with a YAML front matter block between `---` lines, the converter emits a complete HTML5 document with the metadata in `<head>` and the body content between `<body>` tags. Without front matter, the output stays a bare HTML fragment. The `--css` option (or `include_css=True`) embeds the default `<style>` block regardless of whether front matter is present.

```
---
lang: en
title: My Document
author: Jane Doe
description: A short description.
keywords: python, markdown, html5
published: 2026-08-09
---
```

- `lang` becomes the `<html lang="...">` attribute (a valid BCP 47 tag).
- `title` becomes the `<title>` element.
- `author`, `description`, `keywords`, and `published` become `<meta name="..." content="..." />` tags.

- A default `<style>` block with viewing-friendly CSS is embedded in `<head>` .

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="author" content="Jane Doe" />
    <meta name="description" content="A short description." />
    <meta name="keywords" content="python, markdown, html5" />
    <title>My Document</title>
    <meta name="published" content="2026-08-09" />
    <style>
      /* default viewing-friendly CSS */
    </style>
  </head>
  <body>
    ...
  </body>
</html>
```

Any other keys are ignored, and if the file has no front matter at all, the output remains a bare fragment (unless `--css / include_css=True` is used, in which case it becomes a full document).

19. Backslash Escaping

Escape any special char: `\*` , `\#` , `\[` , etc.

Escapable: `\`*_{}[]()#+-.!|~^=:<>`

20. Paragraphs

Consecutive text lines merge into `<p>` . Blank lines separate paragraphs.

Empty line after list close => `<!-- -->` preserves whitespace.

21. CSS Styles

The converter (with `--css` option or `include_css=True`) embeds a default `<style>` block in `<head>` that provides viewing-friendly styling, regardless of YAML front matter. The predefined CSS rules include:

```
body {
  padding: 20px;
  font-family: "Noto Serif", "Liberation Serif", "Times New Roman", Times, serif;
  font-size: 18px;
  line-height: 1.4;
  color: #000000;
  background-color: #ffffff;
}
h1 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 32px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h2 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 28px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h2#toc { font-style: italic; }
h3 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 24px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h4 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
```

```
font-weight: bold;
font-size: 20px;
hyphens: auto;
word-break: normal;
overflow-wrap: break-word;
text-wrap: balance;
}
h5 {
margin-top: 1.2em;
margin-bottom: 0.6em;
font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
font-weight: bold;
font-size: 18px;
hyphens: auto;
word-break: normal;
overflow-wrap: break-word;
text-wrap: balance;
}
h6 {
margin-top: 1.2em;
margin-bottom: 0.6em;
font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
font-weight: bold;
font-size: 18px;
font-style: italic;
hyphens: auto;
word-break: normal;
overflow-wrap: break-word;
text-wrap: balance;
}
p {
hyphens: auto;
hyphenate-limit-chars: 6 3 3;
word-break: normal;
overflow-wrap: break-word;
}
hr {
height: 4px;
margin-top: 32pt;
border: none;
background-color: #000000;
}
blockquote {
margin-left: 0;
padding-left: 20px;
border-left: 8px solid #f5f5f5;
```

```
hyphens: auto;
hyphenate-limit-chars: 6 3 3;
word-break: normal;
overflow-wrap: break-word;
}
mark {
padding: 0 2px;
border-radius: 4px;
background-color: #ffff00;
color: #000000;
}
a:link { color: #0000cd; }
a:visited { color: #9400d3; }
a:hover { outline: none; color: #000080; }
a:focus { outline: none; color: #000080; }
a:active { color: #dc143c; }
ol {
hyphens: auto;
hyphenate-limit-chars: 6 3 3;
word-break: normal;
overflow-wrap: break-word;
}
ul {
hyphens: auto;
hyphenate-limit-chars: 6 3 3;
word-break: normal;
overflow-wrap: break-word;
}
li {
position: relative;
padding-left: 20px;
hyphens: auto;
hyphenate-limit-chars: 6 3 3;
word-break: normal;
overflow-wrap: break-word;
}
ol li input[type="checkbox"], ul li input[type="checkbox"] {
vertical-align: baseline;
accent-color: #000000;
opacity: 1;
cursor: not-allowed;
}
dt {
font-weight: bold;
hyphens: auto;
word-break: break-word;
}
```

```
    overflow-wrap: anywhere;
}
dd {
    position: relative;
    margin-left: 0;
    padding-left: 20px;
    font-style: italic;
    hyphens: auto;
    hyphenate-limit-chars: 6 3 3;
    word-break: normal;
    overflow-wrap: break-word;
}
code {
    padding: 2px 4px;
    border-radius: 4px;
    font-family: "Noto Sans Mono", "Liberation Mono", "Courier New", Courier, monospace;
    font-size: 0.9em;
    line-height: 1;
    hyphens: none !important;
    white-space: normal;
    word-break: break-all;
    overflow-wrap: anywhere;
}
pre {
    max-width: 100%;
    margin: 0;
    padding: 20px;
    border: 1px solid #000000;
    background-color: #f5f5f5;
    overflow: auto;
    scrollbar-color: #000000 transparent;
    white-space: pre-wrap;
    word-break: break-all;
    overflow-wrap: anywhere;
}
pre > code {
    display: block;
    margin: 0;
    padding: 0;
    border: none;
    border-radius: 0;
    line-height: 1.2;
    background-color: transparent;
    overflow: visible;
    hyphens: none !important;
    white-space: pre;
```



```
letter-spacing: 0.05em;
font-size: 0.55em;
line-break: strict;
white-space: nowrap;
overflow-wrap: normal;
}
rp { display: none; }
span[lang="ja"] {
  font-family: "Noto Serif CJK JP", "Source Han Serif JP", "源ノ明朝", "Source Han
Serif", "Hiragino Mincho ProN", "Hiragino Mincho Pro", "IPAexMincho", "IPAMincho", "MS
PMincho", "MS Mincho", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-CN"] {
  font-family: "Noto Serif CJK SC", "Source Han Serif SC", "思源宋体", "Source Han Serif
CN", "Source Han Serif", "Songti SC", "FandolSong", "WenQuanYi Bitmap Song", "SimSun",
serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-Hans"] {
  font-family: "Noto Serif CJK SC", "Source Han Serif SC", "思源宋体", "Source Han Serif
CN", "Source Han Serif", "Songti SC", "FandolSong", "WenQuanYi Bitmap Song", "SimSun",
serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-TW"] {
  font-family: "Noto Serif CJK TC", "Source Han Serif TC", "思源宋體", "Source Han Serif
TW", "Source Han Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "PMingLiU",
"MingLiU", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-Hant"] {
  font-family: "Noto Serif CJK TC", "Source Han Serif TC", "思源宋體", "Source Han Serif
TW", "Source Han Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "PMingLiU",
"MingLiU", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-HK"] {
  font-family: "Noto Serif CJK HK", "Source Han Serif HK", "思源宋體 香港", "思源宋體",
"Source Han Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "MingLiU_HKSCS",
"PMingLiU", "MingLiU", serif;
```

```
word-break: break-all;
line-break: normal;
}
span[lang="ko"] {
  font-family: "Noto Serif CJK KR", "Source Han Serif KR", "본명조", "Source Han Serif",
"AppleMyungjo", "UnBatang", "은바탕", "Batang", serif;
  word-break: break-all;
  line-break: normal;
}
```